



8.2.1 Basics of descent methods ¶

8.2.1 Descent methods, Part 1

fit width



Remark 8.2.1.1. In the video, the quadratic polynomial pictured takes on the value $-\frac{1}{2}\hat{x}^T A \hat{x}$ at $\hat{x}$ and that minimum is below the x-axis. This does not change the conclusions that are drawn in the video.

The basic idea behind a descent method is that at the k th iteration one has an approximation to x , $x^{(k)}$, and one would like to create a better approximation, $x^{(k+1)}$. To do so, the method picks a **search direction**, $p^{(k)}$, and chooses the next approximation by taking a step from the current approximate solution in the direction of $p^{(k)}$:

$$x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}.$$

In other words, one searches for a minimum along a line defined by the current iterate, $x^{(k)}$, and the search direction, $p^{(k)}$. One then picks α_k so that, preferably, $f(x^{(k+1)}) \leq f(x^{(k)})$. This is summarized in [Figure 8.2.1.2](#).

```
Given : A, b, x(0)
r(0) := b - Ax(0)
k := 0
while r(k) ≠ 0
    p(k) := next direction
    x(k+1) := x(k) + αkp(k) for some scalar αk
    r(k+1) := b - Ax(k+1)
    k := k + 1
endwhile
```

Figure 8.2.1.2. Outline for a descent method.
To this goal, typically, an **exact descent method** picks α_k to exactly minimize the function along the line from the current approximate solution in the direction of $p^{(k)}$.

8.2.1 Descent methods, Part 2

fit width



Now,

$$\begin{aligned} f(x^{(k+1)}) &= < x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)} > \\ f(x^{(k)} + \alpha_k p^{(k)}) &= < \text{evaluate} > \\ \frac{1}{2} (x^{(k)} + \alpha_k p^{(k)})^T A (x^{(k)} + \alpha_k p^{(k)}) - (x^{(k)} + \alpha_k p^{(k)})^T b &= < \text{multiply out} > \\ \frac{1}{2} x^{(k)T} A x^{(k)} + \alpha_k p^{(k)T} A x^{(k)} + \frac{1}{2} \alpha_k^2 p^{(k)T} A p^{(k)} - x^{(k)T} b - \alpha_k p^{(k)T} b &= < \text{rearrange} > \\ \frac{1}{2} x^{(k)T} A x^{(k)} - x^{(k)T} b + \frac{1}{2} \alpha_k^2 p^{(k)T} A p^{(k)} + \alpha_k p^{(k)T} A x^{(k)} - \alpha_k p^{(k)T} b &= < \text{substitute } f(x^{(k)}) \text{ and factor out common terms} > \\ f(x^{(k)}) + \frac{1}{2} \alpha_k^2 p^{(k)T} A p^{(k)} + \alpha_k p^{(k)T} (A x^{(k)} - b) &= < \text{substitute } r^{(k)} \text{ and commute to expose polynomial in } \alpha_k > \\ \frac{1}{2} p^{(k)T} A p^{(k)} \alpha_k^2 - p^{(k)T} r^{(k)} \alpha_k + f(x^{(k)}), \end{aligned}$$

where $r^{(k)} = b - A x^{(k)}$ is the **residual**. This is a quadratic polynomial in the scalar α_k (since this is the only free variable).

8.2.1 Descent methods, Part 3

fit width



Minimizing

$$f(x^{(k+1)}) = \frac{1}{2} p^{(k)T} A p^{(k)} \alpha_k^2 - p^{(k)T} r^{(k)} \alpha_k + f(x^{(k)})$$

exactly requires the derivative with respect to α_k to be zero:

$$0 = \frac{df(x^{(k)} + \alpha_k p^{(k)})}{d\alpha_k} = p^{(k)T} A p^{(k)} \alpha_k - p^{(k)T} r^{(k)}.$$

Hence, for a given choice of p_k

$$\alpha_k = \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}} \quad \text{and} \quad x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)}.$$

provides the next approximation to the solution. This leaves us with the question of how to pick the search directions $\{p^{(0)}, p^{(1)}, \dots\}$.

A basic decent method based on these ideas is given in [Figure 8.2.1.3](#).

```

Given :  $A, b, x^{(0)}$ 
 $r^{(0)} := b - Ax^{(0)}$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
   $p^{(k)} :=$  next direction
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := b - Ax^{(k+1)}$ 
   $k := k + 1$ 
endwhile
  
```

Figure 8.2.1.3. Basic descent method.

Homework 8.2.1.1. The cost of an iterative method is a combination of how many iterations it takes to convergence and the cost per iteration. For the loop in [Figure 8.2.1.3](#), count the number of matrix-vector multiplications, dot products, and "axy" operations (not counting the cost of determining the next descent direction).

▼ [Solution](#)

$$\begin{array}{ll} \alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}} & 1 \text{ mvmult, 2 dot products} \\ x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)} & 1 \text{ axpy} \\ r^{(k+1)} := b - Ax^{(k+1)} & 1 \text{ mvmult} \end{array}$$

Total: 2 matrix-vector multiplies (mvmults), 2 dot products, 1 axpy.